

AWS NOTES

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1. Introduction

Problems with traditional IT approach

- Pay for the rent for the data center
- Pay for power supply, cooling, maintenance
- Adding and replacing hardware takes time
- Scaling is limited
- Hire 24/7 team to monitor the infrastructure
- How to deal with disasters? (Earthquake, power shutdown, fire...)

- Can we externalize all this?



What is Cloud Computing?

- Cloud computing is the on-demand delivery of compute power, database storage, applications, and other IT resources
- Through a cloud services platform with pay-as-you-go pricing
- You can provision exactly the right type and size of computing resources you need
- You can access as many resources as you need, almost instantly
- Simple way to access servers, storage, databases and a set of application services
- Amazon Web Services owns and maintains the network-connected hardware required for these application services, while you provision and use what you need via a web application.

You've been using some Cloud services



Gmail

- E-mail cloud service
- Pay for ONLY your emails stored (no infrastructure, etc.)



Dropbox

- Cloud Storage Service
- Originally built on AWS



Netflix

- Built on AWS
- Video on Demand

Six Advantages of Cloud Computing

- Trade capital expense (CAPEX) for operational expense (OPEX)
 - Pay On-Demand: don't own hardware
 - Reduced Total Cost of Ownership (TCO) & Operational Expense (OPEX)
- Benefit from massive economies of scale
 - Prices are reduced as AWS is more efficient due to large scale
- Stop guessing capacity
 - Scale based on actual measured usage
- Increase speed and agility
- Stop spending money running and maintaining data centers
- Go global in minutes: leverage the AWS global infrastructure

Problems solved by the Cloud

- Flexibility: change resource types when needed
- Cost-Effectiveness: pay as you go, for what you use
- Scalability: accommodate larger loads by making hardware stronger or adding additional nodes
- Elasticity: ability to scale out and scale-in when needed
- High-availability and fault-tolerance: build across data centers
- Agility: rapidly develop, test and launch software applications

The Deployment Models of the Cloud

Private Cloud:

- Cloud services used by a single organization, not exposed to the public.
- Complete control
- Security for sensitive applications
- Meet specific business needs



Public Cloud:

- Cloud resources owned and operated by a third-party cloud service provider delivered over the Internet.
- Six Advantages of Cloud Computing



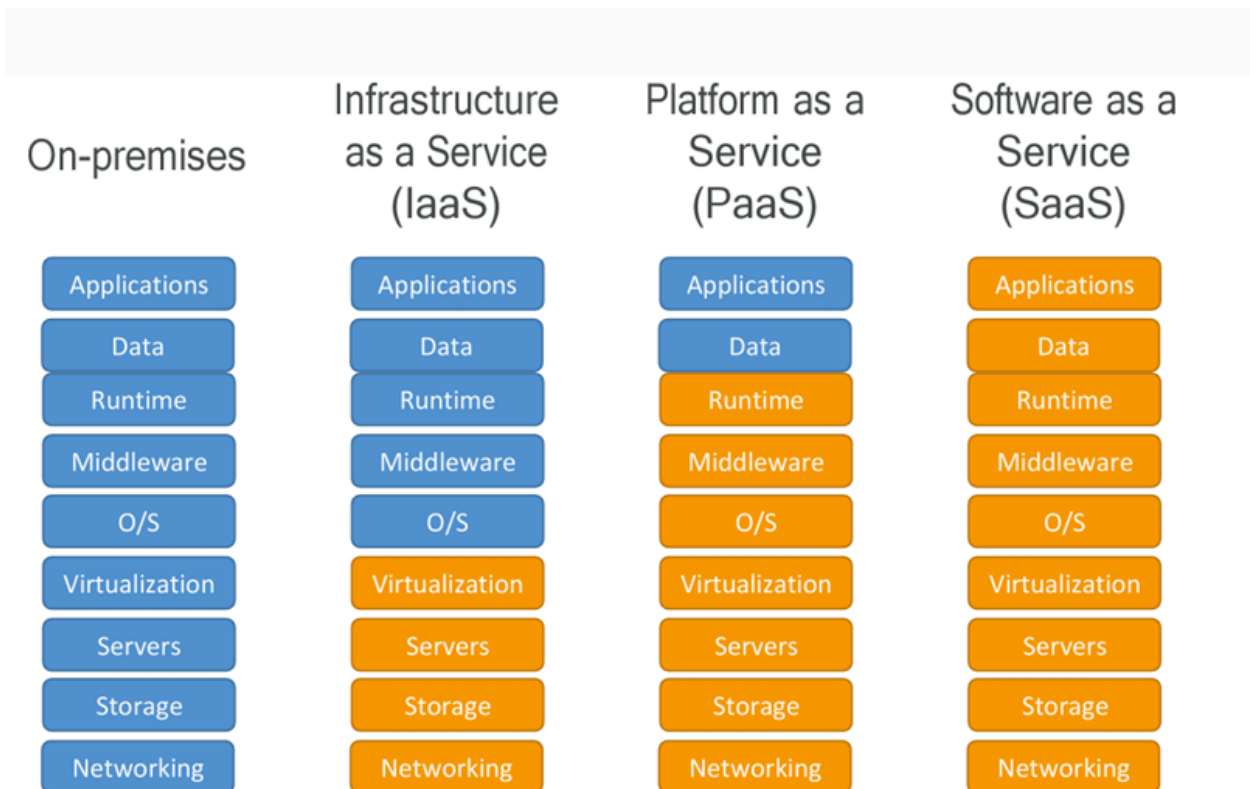
Hybrid Cloud:

- Keep some servers on premises and extend some capabilities to the Cloud
- Control over sensitive assets in your private infrastructure
- Flexibility and cost-effectiveness of the public cloud



Types of Cloud Computing

- Infrastructure as a Service (IaaS)
 - Provide building blocks for cloud IT
 - Provides networking, computers, data storage space
 - Highest level of flexibility
 - Easy parallel with traditional on-premises IT
- Platform as a Service (PaaS)
 - Removes the need for your organization to manage the underlying infrastructure
 - Focus on the deployment and management of your applications
- Software as a Service (SaaS)
 - Completed product that is run and managed by the service provider



Example of Cloud Computing Types

- Infrastructure as a Service:
 - Amazon EC2 (on AWS)
 - GCP, Azure, Rackspace, Digital Ocean, Linode
- Platform as a Service:
 - Elastic Beanstalk (on AWS)
 - Heroku, Google App Engine (GCP), Windows Azure (Microsoft)
- Software as a Service:
 - Many AWS services (ex: Rekognition for Machine Learning)
 - Google Apps (Gmail), Dropbox, Zoom



Pricing of the Cloud – Quick Overview

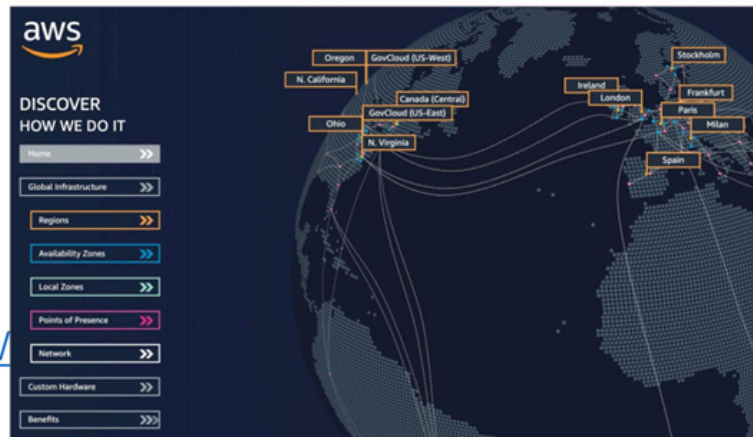
- AWS has 3 pricing fundamentals, following the pay-as-you-go pricing model
- Compute:
 - Pay for compute time
- Storage:
 - Pay for data stored in the Cloud
- Data transfer OUT of the Cloud:
 - Data transfer IN is free
- Solves the expensive issue of traditional IT



AWS Global Infrastructure

- AWS Regions
- AWS Availability Zones
- AWS Data Centers
- AWS Edge Locations / Points of Presence

- <https://infrastructure.aws/>



2. Amazon S3

Amazon S3 - Buckets

- Amazon S3 allows people to store objects (files) in “buckets” (directories)
- Buckets must have a globally unique name (across all regions all accounts)
- Buckets are defined at the region level
- S3 looks like a global service but buckets are created in a region
- Naming convention
 - No uppercase, No underscore
 - 3-63 characters long
 - Not an IP
 - Must start with lowercase letter or number
 - Must NOT start with the prefix xn--
 - Must NOT end with the suffix -s3alias



S3 Bucket

Amazon S3 - Objects

Objects (files) have a Key

The **key** is the FULL path:

- s3://my-bucket/my_file.txt
- s3://my-bucket/my_folder1/another_folder/my_file.txt
- The key is composed of **prefix** + **object name**
 - s3://my-bucket/my_folder1/another_folder/my_file.txt
- There's no concept of "directories" within buckets (although the UI will trick you to think otherwise)
- Just keys with very long names that contain slashes ("/")



Object



S3 Bucket
with Objects

Amazon S3 - Objects (cont.)



- Object values are the content of the body:
 - Max. Object Size is 5TB (5000GB)
 - If uploading more than 5GB, must use "multi-part upload"
- Metadata (list of text key / value pairs - system or user metadata)
- Tags (Unicode key / value pair - up to 10) - useful for security / lifecycle
- Version ID (if versioning is enabled)

S3 Bucket Policies

- JSON based policies
 - Resources: buckets and objects
 - Effect: Allow / Deny
 - Actions: Set of API to Allow or Deny
 - Principal: The account or user to apply the policy to
- Use S3 bucket for policy to:
 - Grant public access to the bucket
 - Force objects to be encrypted at upload
 - Grant access to another account (Cross Account)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicRead",
      "Effect": "Allow",
      "Principal": "*",
      "Action": [
        "s3:GetObject"
      ],
      "Resource": [
        "arn:aws:s3:::examplebucket/*"
      ]
    }
  ]
}
```

Amazon S3 - Static Website Hosting

- S3 can host static websites and have them accessible on the Internet

<http://demo-bucket.s3-website-us-west-2.amazonaws.com>
<http://demo-bucket.s3-website.us-west-2.amazonaws.com>

The website URL will be (depending on the region)

- <http://bucket-name.s3-website-aws-region.amazonaws.com>

OR

- <http://bucket-name.s3-website.aws-region.amazonaws.com>

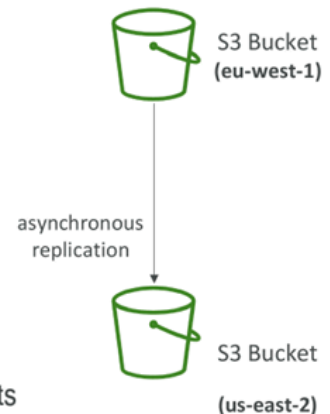
- If you get a 403 Forbidden error, make sure the bucket policy allows public reads!



Amazon S3 - Replication (CRR & SRR)



- Must enable Versioning in source and destination buckets
- Cross-Region Replication (CRR)
- Same-Region Replication (SRR)
- Buckets can be in different AWS accounts
- Copying is asynchronous
- Must give proper IAM permissions to S3
- Use cases:
 - CRR - compliance, lower latency access, replication across accounts
 - SRR - log aggregation, live replication between production and test accounts



S3 Storage Classes

- Amazon S3 Standard - General Purpose
- Amazon S3 Standard-Infrequent Access (IA)
- Amazon S3 One Zone-Infrequent Access
- Amazon S3 Glacier Instant Retrieval
- Amazon S3 Glacier Flexible Retrieval
- Amazon S3 Glacier Deep Archive
- Amazon S3 Intelligent Tiering
- Can move between classes manually or using S3 Lifecycle configurations

S3 Durability and Availability

- Durability:
 - High durability (99.999999999%, 11 9's) of objects across multiple AZ
 - If you store 10,000,000 objects with Amazon S3, you can on average expect to incur a loss of a single object once every 10,000 years
 - Same for all storage classes
- Availability:
 - Measures how readily available a service is
 - Varies depending on storage class
 - Example: S3 standard has 99.99% availability = not available 53 minutes a year

S3 Storage Classes - Infrequent Access

- For data that is less frequently accessed, but requires rapid access when needed
- Lower cost than S3 Standard

- Amazon S3 Standard-Infrequent Access (S3 Standard-IA)

- 99.9% Availability
- Use cases: Disaster Recovery, backups



- Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)

- High durability (99.999999999%) in a single AZ; data lost when AZ is destroyed
- 99.5% Availability
- Use Cases: Storing secondary backup copies of on-premise data, or data you can recreate



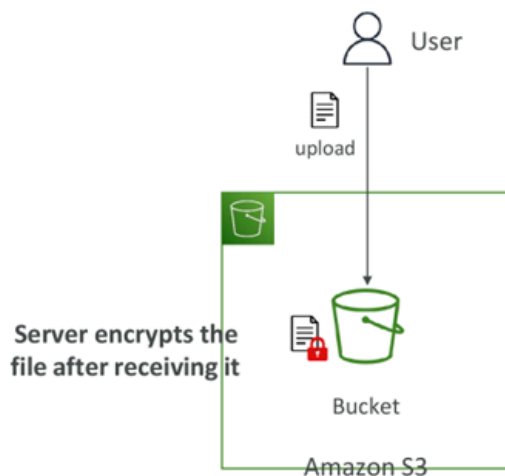
Amazon S3 Glacier Storage Classes

- Low-cost object storage meant for archiving / backup
- Pricing: price for storage + object retrieval cost
- Amazon S3 Glacier Instant Retrieval
 - Millisecond retrieval, great for data accessed once a quarter
 - Minimum storage duration of 90 days
- Amazon S3 Glacier Flexible Retrieval (formerly Amazon S3 Glacier):
 - Expedited (1 to 5 minutes), Standard (3 to 5 hours), Bulk (5 to 12 hours) - free
 - Minimum storage duration of 90 days
- Amazon S3 Glacier Deep Archive - for long term storage:
 - Standard (12 hours), Bulk (48 hours)
 - Minimum storage duration of 180 days

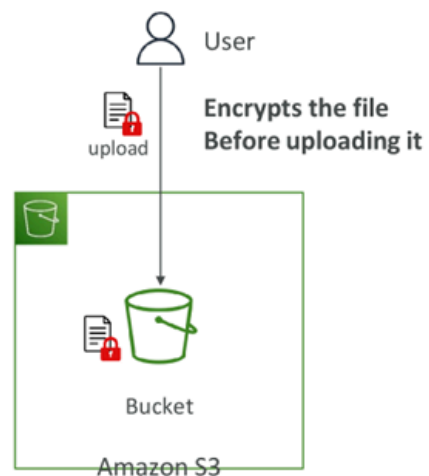


S3 Encryption

Server-Side Encryption (Default)



Client-Side Encryption



Amazon S3 - Summary

- Buckets vs Objects: global unique name, tied to a region
- S3 security: IAM policy, S3 Bucket Policy (public access), S3 Encryption
- S3 Websites: host a static website on Amazon S3
- S3 Versioning: multiple versions for files, prevent accidental deletes
- S3 Replication: same-region or cross-region, must enable versioning
- S3 Storage Classes: Standard, IA, 1Z-IA, Intelligent, Glacier (Instant, Flexible, Deep)

3. IAM

IAM: Users & Groups



- IAM = Identity and Access Management, Global service
- Root account created by default, shouldn't be used or shared
- Users are people within your organization, and can be grouped
- Groups only contain users, not other groups
- Users don't have to belong to a group, and user can belong to multiple groups



IAM Policies Structure

- Consists of
 - Version: policy language version, always include "2012-10-17"
 - Id: an identifier for the policy (optional)
 - Statement: one or more individual statements (required)
- Statements consists of
 - Sid: an identifier for the statement (optional)
 - Effect: whether the statement allows or denies access (Allow, Deny)
 - Principal: account/user/role to which this policy applied to
 - Action: list of actions this policy allows or denies
 - Resource: list of resources to which the actions applied to
 - Condition: conditions for when this policy is in effect (optional)

```
{
  "Version": "2012-10-17",
  "Id": "S3-Account-Permissions",
  "Statement": [
    {
      "Sid": "1",
      "Effect": "Allow",
      "Principal": {
        "AWS": ["arn:aws:iam::123456789012:root"]
      },
      "Action": [
        "s3:GetObject",
        "s3:PutObject"
      ],
      "Resource": ["arn:aws:s3:::mybucket/*"]
    }
  ]
}
```

IAM - Password Policy

- Strong passwords = higher security for your account
- In AWS, you can setup a password policy:
 - Set a minimum password length
 - Require specific character types:
 - including uppercase letters
 - lowercase letters
 - numbers
 - non-alphanumeric characters
 - Allow all IAM users to change their own passwords
 - Require users to change their password after some time (password expiration)
 - Prevent password re-use

Multi Factor Authentication - MFA



- Users have access to your account and can possibly change configurations or delete resources in your AWS account
- You want to protect your Root Accounts and IAM users
- MFA = password *you know* + security device *you own*



Alice

Password

+



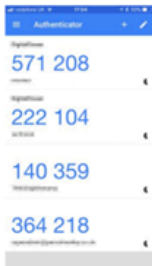
=>

Successful login

- Main benefit of MFA:
if a password is stolen or hacked, the account is not compromised

MFA devices options in AWS

Virtual MFA device



Google Authenticator
(phone only)



Authy
(multi-device)

Support for multiple tokens on a single device.

Universal 2nd Factor (U2F) Security Key



YubiKey by Yubico (3rd party)

Support for multiple root and IAM users
using a single security key

How can users access AWS ?



- To access AWS, you have three options:
 - AWS Management Console (protected by password + MFA)
 - AWS Command Line Interface (CLI): protected by access keys
 - AWS Software Developer Kit (SDK) - for code: protected by access keys
- Access Keys are generated through the AWS Console
- Users manage their own access keys
- Access Keys are secret, just like a password. Don't share them
- Access Key ID ~= username
- Secret Access Key ~= password

What's the AWS CLI?

- A tool that enables you to interact with AWS services using commands in your command-line shell
- Direct access to the public APIs of AWS services
- You can develop scripts to manage your resources
- It's open-source <https://github.com/aws/aws-cli>
- Alternative to using AWS Management Console

```
→ ~ aws s3 cp myfile.txt s3://ccp-mybucket/myfile.txt
upload: ./myfile.txt to s3://ccp-mybucket/myfile.txt
→ ~ aws s3 ls s3://ccp-mybucket
2021-05-14 03:22:52          0 myfile.txt
→ ~
```

IAM Guidelines & Best Practices



- Don't use the root account except for AWS account setup
- One physical user = One AWS user
- Assign users to groups and assign permissions to groups
- Create a strong password policy
- Use and enforce the use of Multi Factor Authentication (MFA)
- Create and use Roles for giving permissions to AWS services
- Use Access Keys for Programmatic Access (CLI / SDK)
- Audit permissions of your account using IAM Credentials Report & IAM Access Advisor

IAM Section - Summary



- Users: mapped to a physical user, has a password for AWS Console
- Groups: contains users only
- Policies: JSON document that outlines permissions for users or groups
- Roles: for EC2 instances or AWS services
- Security: MFA + Password Policy
- AWS CLI: manage your AWS services using the command-line
- AWS SDK: manage your AWS services using a programming language
- Access Keys: access AWS using the CLI or SDK

4. EC2

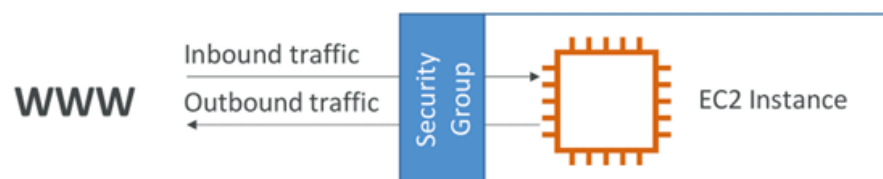
Amazon EC2



- EC2 is one of the most popular of AWS' offerings
- EC2 = Elastic Compute Cloud = Infrastructure as a Service
- It mainly consists in the capability of :
 - Renting virtual machines (EC2)
 - Storing data on virtual drives (EBS)
 - Distributing load across machines (ELB)
 - Scaling the services using an auto-scaling group (ASG)
- Knowing EC2 is fundamental to understand how the Cloud works

Introduction to Security Groups

- Security Groups are the fundamental of network security in AWS
- They control how traffic is allowed into or out of our EC2 Instances.



- Security groups only contain **allow** rules
- Security groups rules can reference by IP or by security group

Classic Ports to know

- 22 = SSH (Secure Shell) - log into a Linux instance
- 21 = FTP (File Transfer Protocol) - upload files into a file share
- 22 = SFTP (Secure File Transfer Protocol) - upload files using SSH
- 80 = HTTP - access unsecured websites
- 443 = HTTPS - access secured websites
- 3389 = RDP (Remote Desktop Protocol) - log into a Windows instance

EC2 Instance Connect

- Connect to your EC2 instance within your browser
- No need to use your key file that was downloaded
- The “magic” is that a temporary key is uploaded onto EC2 by AWS
- Works only out-of-the-box with Amazon Linux 2
- Need to make sure the port 22 is still opened!

Gib vs GB

one GB is defined as 1000^3 (1,000,000,000) bytes and one GiB as 1024^3 (1,073,741,824) bytes. That means one GB equals 0.93 GiB

GB=1000mb(gigabytes)

GiB= 1024(gibibytes)

EC2 Instances Purchasing Options

- On-Demand Instances - short workload, predictable pricing, pay by second
- Reserved (1 & 3 years)
 - Reserved Instances - long workloads
 - Convertible Reserved Instances - long workloads with flexible instances
- Savings Plans (1 & 3 years) –commitment to an amount of usage, long workload
- Spot Instances - short workloads, cheap, can lose instances (less reliable)
- Dedicated Hosts - book an entire physical server, control instance placement
- Dedicated Instances - no other customers will share your hardware
- Capacity Reservations - reserve capacity in a specific AZ for any duration

Which purchasing option is right for me?



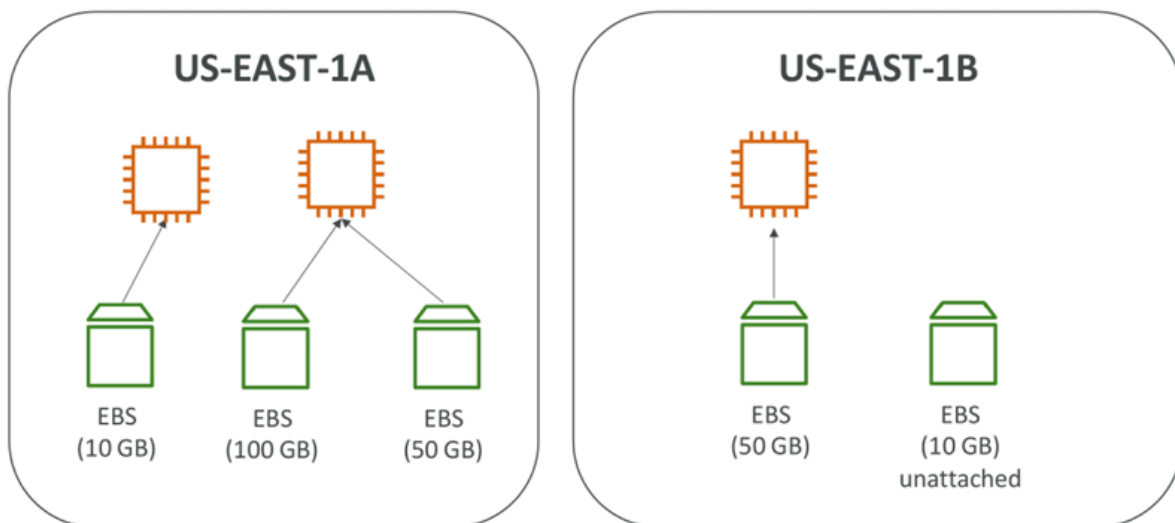
- On demand: coming and staying in resort whenever we like, we pay the full price
- Reserved: like planning ahead and if we plan to stay for a long time, we may get a good discount.
- Savings Plans: pay a certain amount per hour for certain period and stay in any room type (e.g., King, Suite, Sea View, ...)
- Spot instances: the hotel allows people to bid for the empty rooms and the highest bidder keeps the rooms. You can get kicked out at any time
- Dedicated Hosts: We book an entire building of the resort
- Capacity Reservations: you book a room for a period with full price even you don't stay in it

What's an EBS Volume?



- An **EBS (Elastic Block Store) Volume** is a **network** drive you can attach to your instances while they run
 - It allows your instances to persist data, even after their termination
 - They can only be mounted to one instance at a time (at the CCP level)
 - They are bound to a specific availability zone
-
- Analogy: Think of them as a “network USB stick”
 - Free tier: 30 GB of free EBS storage of type General Purpose (SSD) or Magnetic per month

EBS Volume - Example



EBS Snapshots

- Make a backup (snapshot) of your EBS volume at a point in time
- Not necessary to detach volume to do snapshot, but recommended
- Can copy snapshots across AZ or Region



AMI Overview



- AMI = Amazon Machine Image
- AMI are a customization of an EC2 instance
 - You add your own software, configuration, operating system, monitoring...
 - Faster boot / configuration time because all your software is pre-packaged
- AMI are built for a specific region (and can be copied across regions)
- You can launch EC2 instances from:
 - A Public AMI: AWS provided
 - Your own AMI: you make and maintain them yourself
 - An AWS Marketplace AMI: an AMI someone else made (and potentially sells)

EC2 Section - Summary



- EC2 Instance: AMI (OS) + Instance Size (CPU + RAM) + Storage + security groups + EC2 User Data
- Security Groups: Firewall attached to the EC2 instance
- EC2 User Data: Script launched at the first start of an instance
- SSH: start a terminal into our EC2 Instances (port 22)
- EC2 Instance Role: link to IAM roles
- Purchasing Options: On-Demand, Spot, Reserved (Standard + Convertible + Scheduled), Dedicated Host, Dedicated Instance

5. VPC

Before vpc we will see basic of networking

IP Addresses in AWS

- IPv4 - Internet Protocol version 4 (4.3 Billion Addresses)
 - Public IPv4 - can be used on the Internet
 - EC2 instance gets a new a public IP address every time you stop then start it (default)
 - Private IPv4 - can be used on private networks (LAN) such as internal AWS networking (e.g., 192.168.1.1)
 - Private IPv4 is fixed for EC2 Instances even if you start/stop them
- Elastic IP - allows you to attach a fixed public IPv4 address to EC2 instance
 - • Note: has ongoing cost if not attached to EC2 instance or if the EC2 instance is stopped
- IPv6 - Internet Protocol version 6 (3.4×10^{38} Addresses)
 - Every IP address is public (no private range)
 - Example: 2001:db8:3333:4444:cccc:dddd:eeee:ffff

DHCP(dynamic host configuration protol)

Automatically assing ip address when they connect to internet

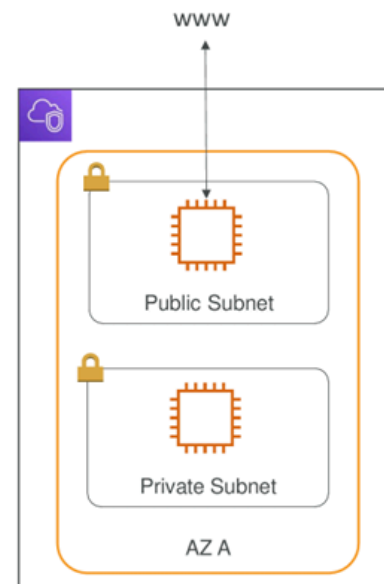
Classes of public ip

Address Class	RANGE	Default Subnet Mask
A	1.0.0.0 to 126.255.255.255	255.0.0.0
B	128.0.0.0 to 191.255.255.255	255.255.0.0
C	192.0.0.0 to 223.255.255.255	255.255.255.0
D	224.0.0.0 to 239.255.255.255	Reserved for Multicasting
E	240.0.0.0 to 254.255.255.255	Experimental

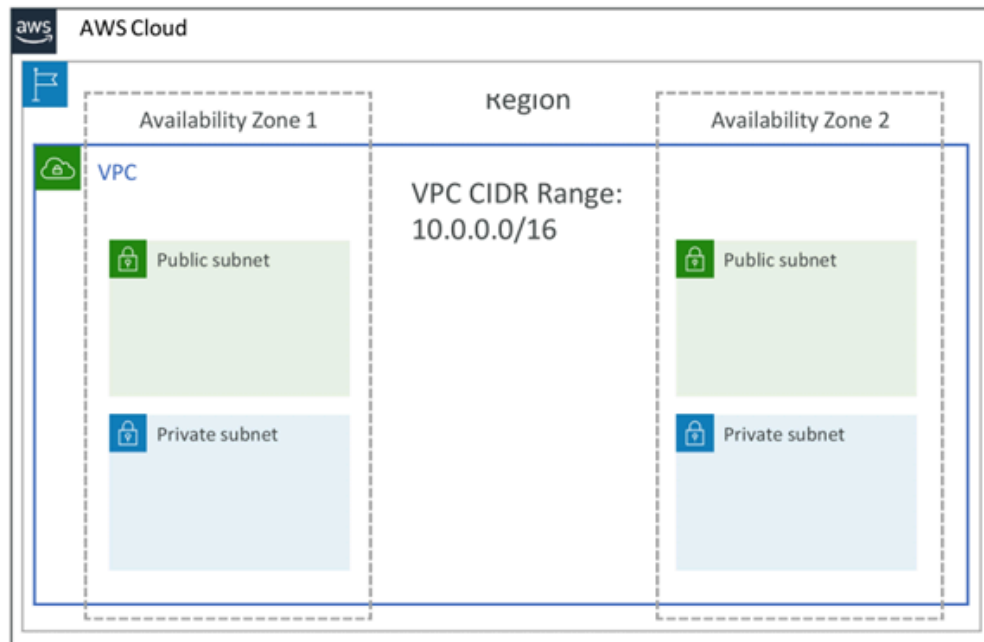
Note: Class A addresses 127.0.0.0 to 127.255.255.255 cannot be used and is reserved for loopback testing.

VPC & Subnets Primer

- VPC - Virtual Private Cloud: private network to deploy your resources (regional resource)
- Subnets allow you to partition your network inside your VPC (Availability Zone resource)
- A public subnet is a subnet that is accessible from the internet
- A private subnet is a subnet that is not accessible from the internet
- To define access to the internet and between subnets, we use Route Tables.

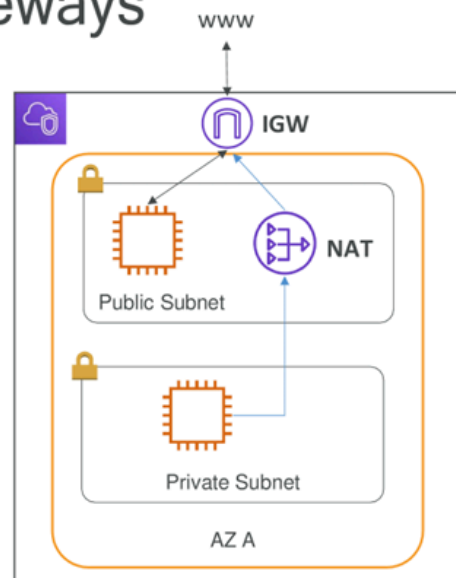


VPC Diagram



Internet Gateway & NAT Gateways

- Internet Gateways helps our VPC instances connect with the internet
- Public Subnets have a route to the internet gateway.
- NAT Gateways (AWS-managed) & NAT Instances (self-managed) allow your instances in your Private Subnets to access the internet while remaining private



Network ACL & Security Groups

- NACL (Network ACL)

- A firewall which controls traffic from and to subnet
- Can have ALLOW and DENY rules
- Are attached at the Subnet level
- Rules only include IP addresses

VPC

Public subnet

NACL

- Security Groups

- A firewall that controls traffic to and from an ENI / an EC2 Instance
- Can have only ALLOW rules
- Rules include IP addresses and other security groups

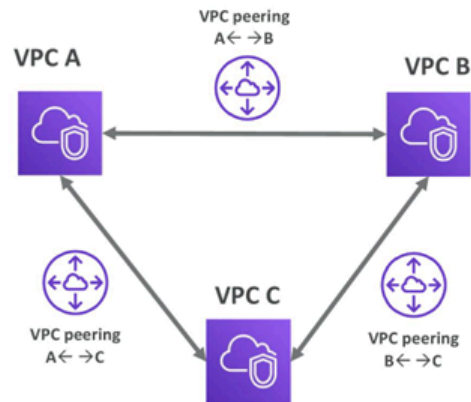
Security group

Network ACLs vs Security Groups

Security Group	Network ACL
Operates at the instance level	Operates at the subnet level
Supports allow rules only	Supports allow rules and deny rules
Is stateful: Return traffic is automatically allowed, regardless of any rules	Is stateless: Return traffic must be explicitly allowed by rules
We evaluate all rules before deciding whether to allow traffic	We process rules in number order when deciding whether to allow traffic
Applies to an instance only if someone specifies the security group when launching the instance, or associates the security group with the instance later on	Automatically applies to all instances in the subnets it's associated with (therefore, you don't have to rely on users to specify the security group)

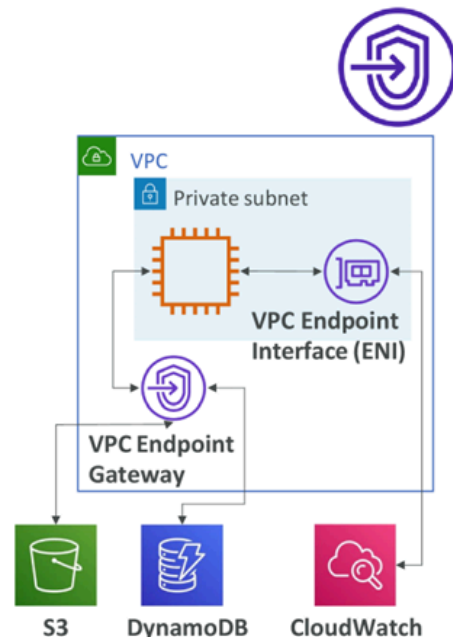
VPC Peering

- Connect two VPC, privately using AWS' network
- Make them behave as if they were in the same network
- Must not have overlapping CIDR (IP address range)
- VPC Peering connection is not transitive (must be established for each VPC that need to communicate with one another)



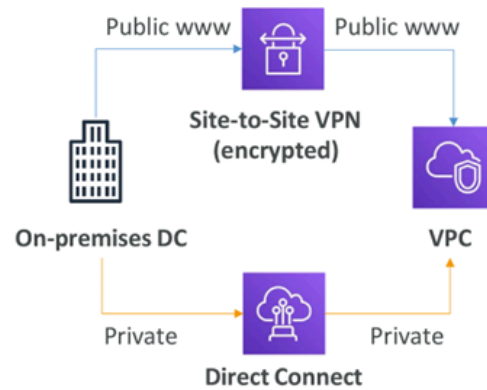
VPC Endpoints

- Endpoints allow you to connect to AWS Services using a private network instead of the public www network
- This gives you enhanced security and lower latency to access AWS services
- VPC Endpoint Gateway: S3 & DynamoDB
- VPC Endpoint Interface: the rest



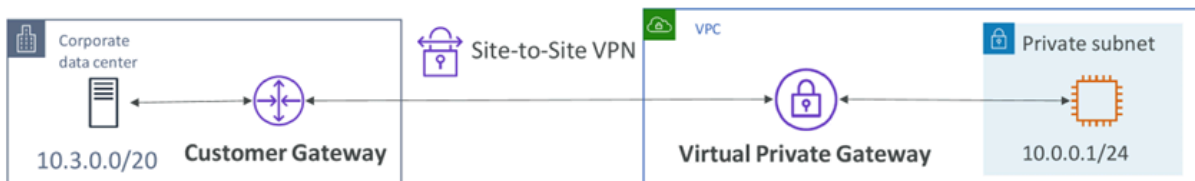
Site to Site VPN & Direct Connect

- Site to Site VPN
 - Connect an on-premises VPN to AWS
 - The connection is automatically encrypted
 - Goes over the public internet
- Direct Connect (DX)
 - Establish a physical connection between on-premises and AWS
 - The connection is private, secure and fast
 - Goes over a private network
 - Takes at least a month to establish



Site-to-Site VPN

- On-premises: must use a Customer Gateway (CGW)
- AWS: must use a Virtual Private Gateway (VGW)



AWS Client VPN



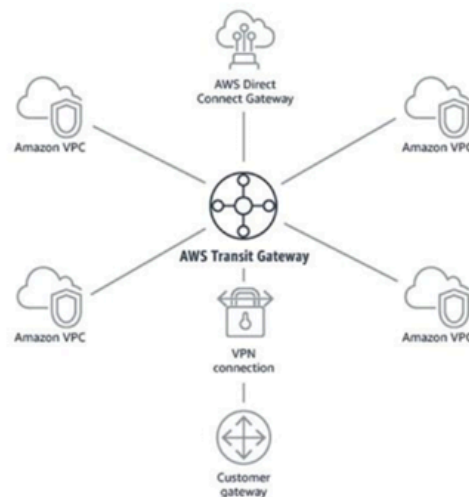
- Connect from your computer using OpenVPN to your private network in AWS and on-premises
- Allow you to connect to your EC2 instances over a private IP (just as if you were in the private VPC network)
- Goes over public Internet



Transit Gateway



- For having transitive peering between thousands of VPC and on-premises, hub-and-spoke (star) connection
- One single Gateway to provide this functionality
- Works with Direct Connect Gateway, VPN connections



VPC Closing Comments

- VPC - Virtual Private Cloud
- Subnets - Tied to an AZ, network partition of the VPC
- Internet Gateway - at the VPC level, provide Internet Access
- NAT Gateway / Instances - give internet access to private subnets
- NACL - Stateless, subnet rules for inbound and outbound
- Security Groups - Stateful, operate at the EC2 instance level or ENI
- VPC Peering - Connect two VPC with non overlapping IP ranges, nontransitive
- Elastic IP –fixed public IPv4, ongoing cost if not in-use

6. ELB, ASG and CloudWatch

Scalability & High Availability

- Scalability means that an application / system can handle greater loads by adapting.
- There are two kinds of scalability:
 - Vertical Scalability
 - Horizontal Scalability (= elasticity)
- Scalability is linked but different to High Availability
- Let's deep dive into the distinction, using a call center as an example

High Availability & Scalability For EC2

- Vertical Scaling: Increase instance size (= scale up / down)
 - From: t2.nano - 0.5G of RAM, 1 vCPU
 - To: u-12tb1.metal - 12.3 TB of RAM, 448 vCPUs
- Horizontal Scaling: Increase number of instances (= scale out / in)
 - Auto Scaling Group
 - Load Balancer
- High Availability: Run instances for the same application across multi AZ
 - Auto Scaling Group multi AZ
 - Load Balancer multi AZ

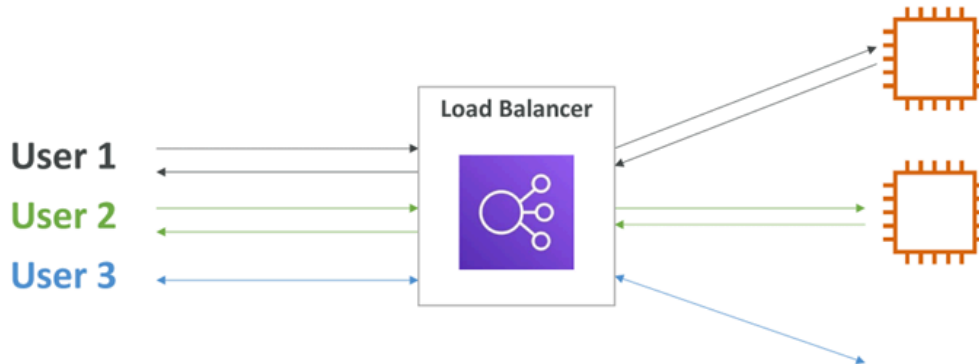
Scalability vs Elasticity (vs Agility)

- Scalability: ability to accommodate a larger load by making the hardware stronger (scale up), or by adding nodes (scale out)
- Elasticity: once a system is scalable, elasticity means that there will be some “auto-scaling” so that the system can scale based on the load. This is “cloud-friendly”: pay-per-use, match demand, optimize costs
- Agility: (not related to scalability - distractor) new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes.

What is load balancing?



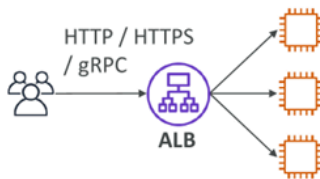
- Load balancers are servers that forward internet traffic to multiple servers (EC2 Instances) downstream.



Application Load Balancer



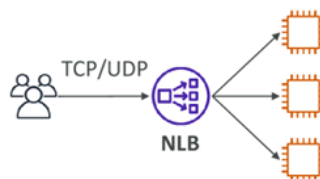
- HTTP / HTTPS / gRPC protocols (Layer 7)
- HTTP Routing features
- Static DNS (URL)



Network Load Balancer



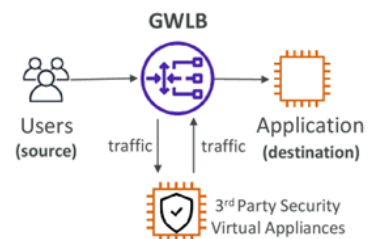
- TCP / UDP protocols (Layer 4)
- High Performance: millions of request per seconds
- Static IP through Elastic IP



Gateway Load Balancer



- GENEVE Protocol on IP Packets (Layer 3)
- Route Traffic to Firewalls that you manage on EC2 Instances
- Intrusion detection

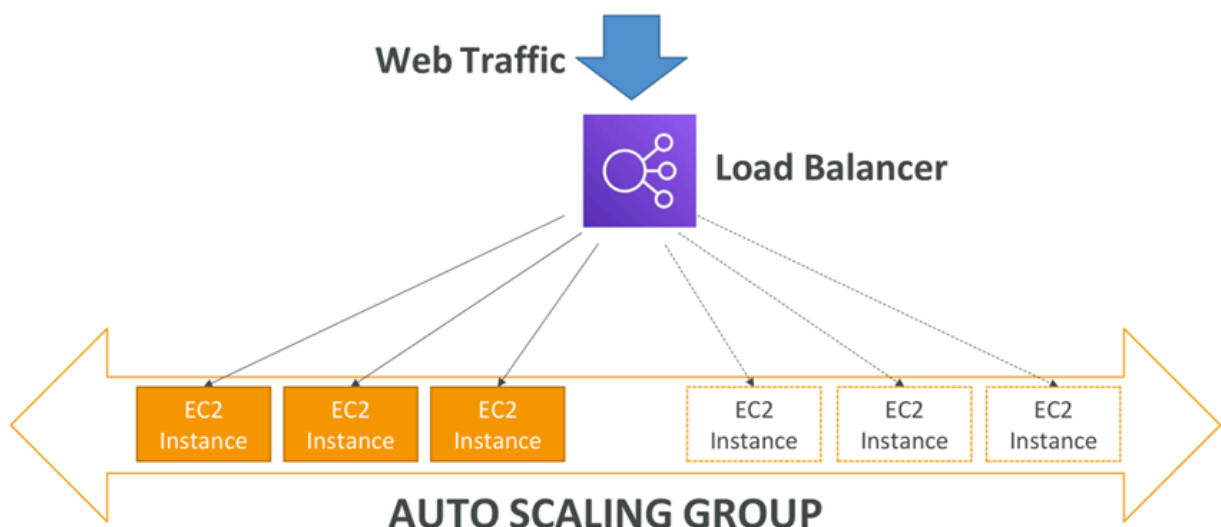


What's an Auto Scaling Group?



- In real-life, the load on your websites and application can change
- In the cloud, you can create and get rid of servers very quickly
- The goal of an Auto Scaling Group (ASG) is to:
 - Scale out (add EC2 instances) to match an increased load
 - Scale in (remove EC2 instances) to match a decreased load
 - Ensure we have a minimum and a maximum number of machines running
 - Automatically register new instances to a load balancer
 - Replace unhealthy instances
- Cost Savings: only run at an optimal capacity (principle of the cloud)

Auto Scaling Group in AWS With Load Balancer



Amazon CloudWatch Metrics



- CloudWatch provides metrics for every services in AWS
- Metric is a variable to monitor (CPUUtilization, NetworkIn...)
- Metrics have timestamps
- Can create CloudWatch dashboards of metrics

Important Metrics

- EC2 instances: CPU Utilization, Status Checks, Network (not RAM)
 - Default metrics every 5 minutes
 - Option for Detailed Monitoring (\$\$\$): metrics every 1 minute
- EBS volumes: Disk Read/Writes
- S3 buckets: BucketSizeBytes, NumberOfObjects, AllRequests
- Billing: Total Estimated Charge (only in us-east-1)
- Service Limits: how much you've been using a service API
- Custom metrics: push your own metrics

Amazon CloudWatch Alarms



- Alarms are used to trigger notifications for any metric
- Alarms actions...
 - Auto Scaling: increase or decrease EC2 instances “desired” count
 - EC2 Actions: stop, terminate, reboot or recover an EC2 instance
 - SNS notifications: send a notification into an SNS topic
- Various options (sampling, %, max, min, etc...)
- Can choose the period on which to evaluate an alarm
- Example: create a billing alarm on the CloudWatch Billing metric
- Alarm States: OK, INSUFFICIENT_DATA, ALARM

ELB & ASG - Summary

- High Availability vs Scalability (vertical and horizontal) vs Elasticity vs Agility in the Cloud
- Elastic Load Balancers (ELB)
 - Distribute traffic across backend EC2 instances, can be Multi-AZ
 - Supports health checks
 - 4 types: Classic (old), Application (HTTP - L7), Network (TCP - L4), Gateway (L3)
- Auto Scaling Groups (ASG)
 - Implement Elasticity for your application, across multiple AZ
 - Scale EC2 instances based on the demand on your system, replace unhealthy
 - Integrated with the ELB
- Cloud watch :help to monitor aws services

7. Database

Databases Intro



- Storing data on disk (EFS, EBS, EC2 Instance Store, S3) can have its limits
 - Sometimes, you want to store data in a database...
 - You can structure the data
 - You build indexes to efficiently query / search through the data
 - You define relationships between your datasets
-
- Databases are optimized for a purpose and come with different features, shapes and constraints

Relational Databases

- Looks just like Excel spreadsheets, with links between them!
- Can use the SQL language to perform queries / lookups



NoSQL Databases

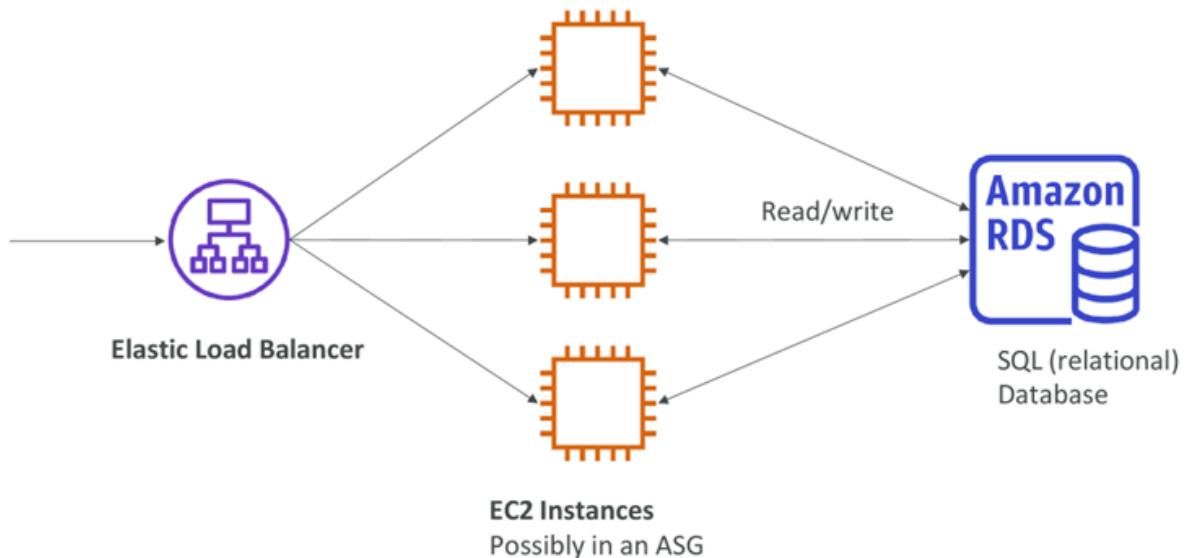
- NoSQL = non-SQL = non relational databases
- NoSQL databases are purpose built for specific data models and have flexible schemas for building modern applications.
- Benefits:
 - Flexibility: easy to evolve data model
 - Scalability: designed to scale-out by using distributed clusters
 - High-performance: optimized for a specific data model
 - Highly functional: types optimized for the data model
- Examples: Key-value, document, graph, in-memory, search databases

AWS RDS Overview



- RDS stands for Relational Database Service
- It's a managed DB service for DB use SQL as a query language.
- It allows you to create databases in the cloud that are managed by AWS
 - Postgres
 - MySQL
 - MariaDB
 - Oracle
 - Microsoft SQL Server
 - Aurora (AWS Proprietary database)

RDS Solution Architecture



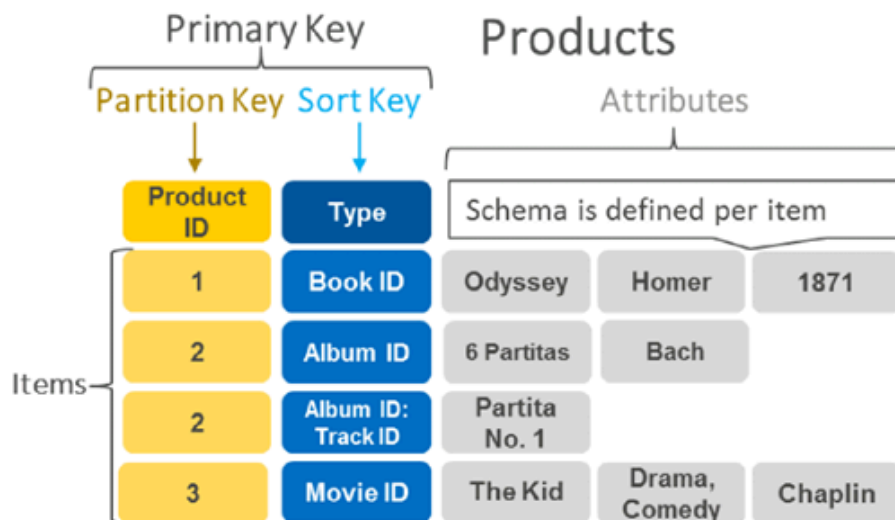
DynamoDB



- Fully Managed Highly available with replication across 3 AZ
- NoSQL database - not a relational database
- Scales to massive workloads, distributed “serverless” database
- Millions of requests per seconds, trillions of row, 100s of TB of storage
- Fast and consistent in performance
- Single-digit millisecond latency - low latency retrieval
- Integrated with IAM for security, authorization and administration
- Low cost and auto scaling capabilities
- Standard & Infrequent Access (IA) Table Class

DynamoDB - type of data

- DynamoDB is a key/value database



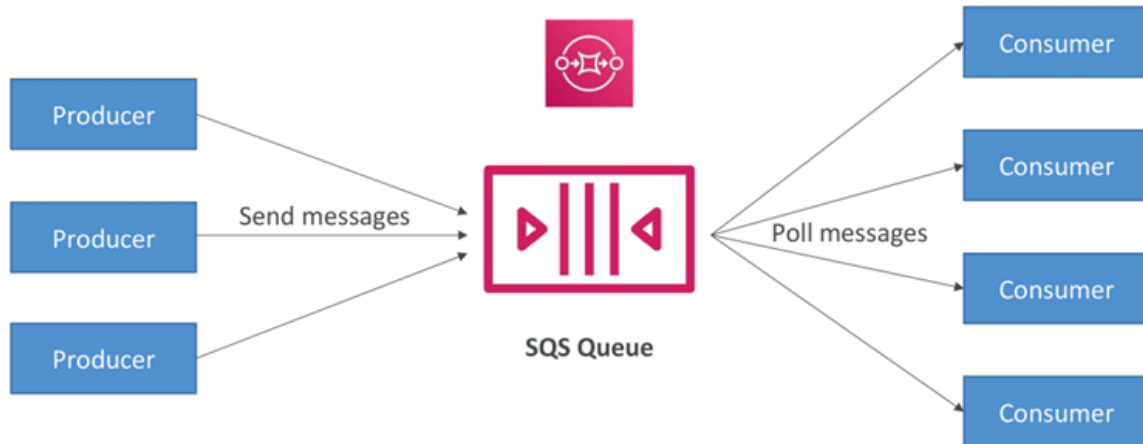
Redshift Overview



- Redshift is based on PostgreSQL, but it's not used for OLTP
- It's OLAP - online analytical processing (analytics and data warehousing)
- Load data once every hour, not every second
- 10x better performance than other data warehouses, scale to PBs of data
- Columnar storage of data (instead of row based)
- Massively Parallel Query Execution (MPP), highly available
- Pay as you go based on the instances provisioned
- Has a SQL interface for performing the queries
- BI tools such as AWS Quicksight or Tableau integrate with it

8. SQS, SWF, SNS

Amazon SQS - Simple Queue Service What's a queue?



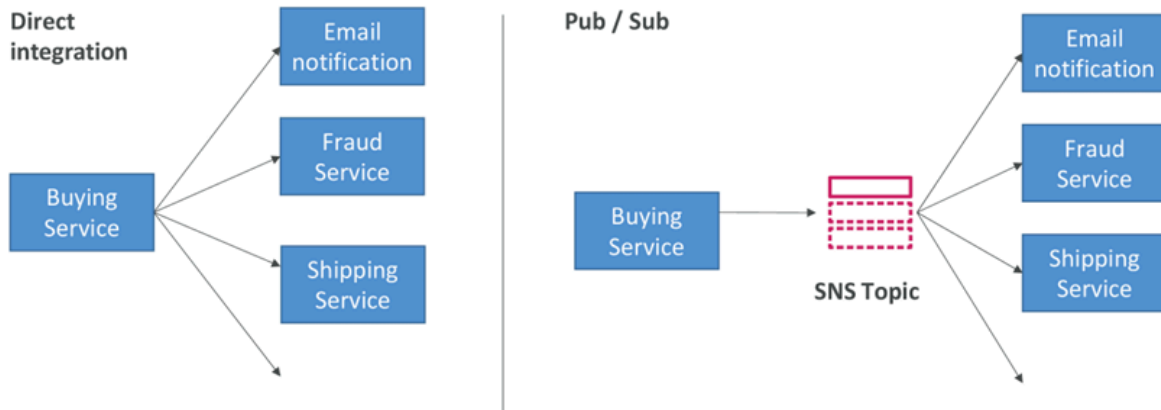
Amazon SQS - Standard Queue



- Oldest AWS offering (over 10 years old)
- Fully managed service (~serverless), use to decouple applications
- Scales from 1 message per second to 10,000s per second
- Default retention of messages: 4 days, maximum of 14 days
- No limit to how many messages can be in the queue
- Messages are deleted after they're read by consumers
- Low latency (<10 ms on publish and receive)
- Consumers share the work to read messages & scale horizontally

Amazon SNS

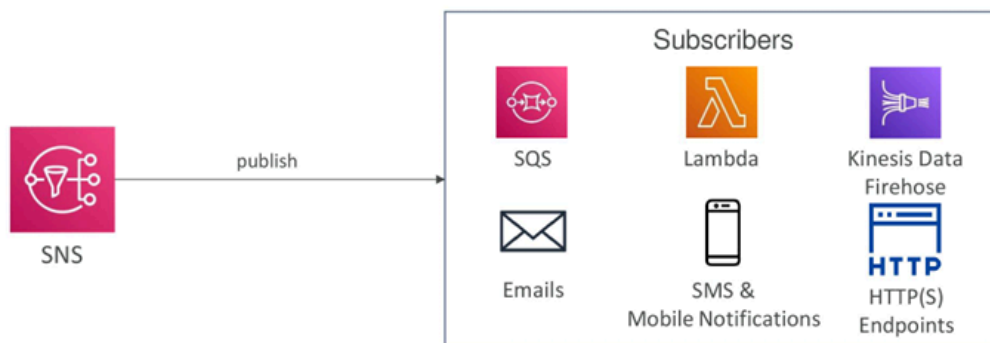
- What if you want to send one message to many receivers?



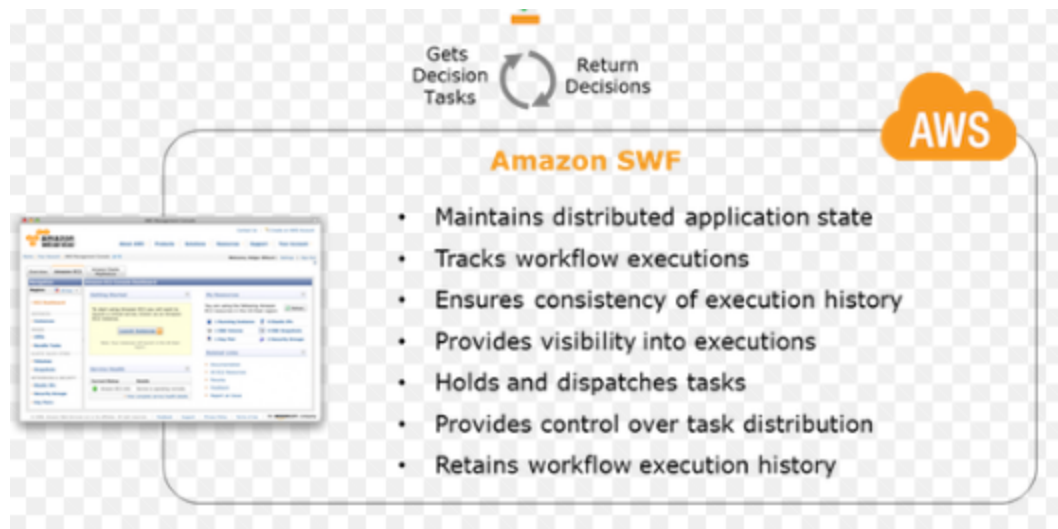
Amazon SNS



- The “event publishers” only sends message to one SNS topic
- As many “event subscribers” as we want to listen to the SNS topic notifications
- Each subscriber to the topic will get all the messages
- Up to 12,500,000 subscriptions per topic, 100,000 topics limit

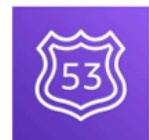


Amazon SWF



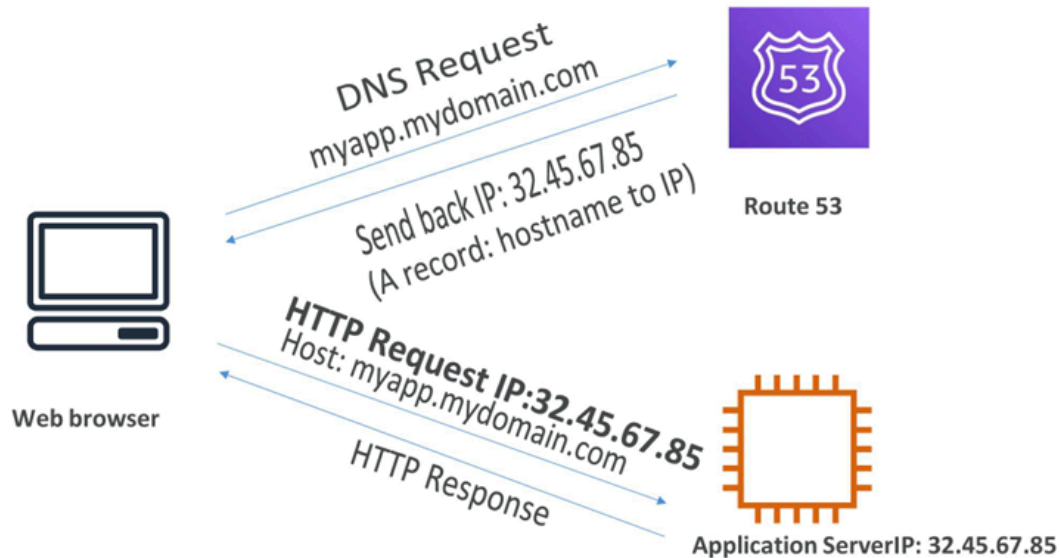
9. DNS and Route 53

Amazon Route 53 Overview



- Route53 is a Managed DNS (Domain Name System)
- DNS is a collection of rules and records which helps clients understand how to reach a server through URLs.
- In AWS, the most common records are:
 - `www.google.com => 12.34.56.78 == A record (IPv4)`
 - `www.google.com => 2001:0db8:85a3:0000:0000:8a2e:0370:7334 == AAAA IPv6`
 - `search.google.com => www.google.com == CNAME: hostname to hostname`
 - `example.com => AWS resource == Alias (ex: ELB, CloudFront, S3, RDS, etc...)`

Route 53 - Diagram for A Record



10. Elastic cache

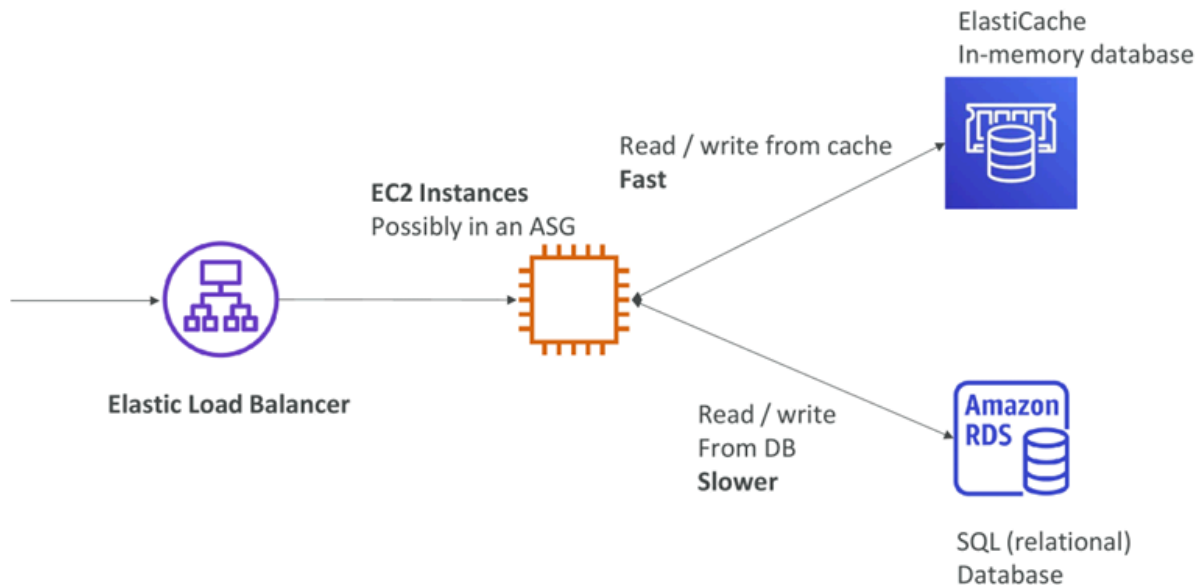
Amazon ElastiCache Overview



- The same way RDS is to get managed Relational Databases...
- ElastiCache is to get managed Redis or Memcached
- Caches are in-memory databases with high performance, low latency
- Helps reduce load off databases for read intensive workloads
- AWS takes care of OS maintenance / patching, optimizations, setup, configuration, monitoring, failure recovery and backups

ElastiCache

Solution Architecture - Cache



11. Key Services(server less)

Why AWS Lambda



Amazon EC2

- Virtual Servers in the Cloud
- Limited by RAM and CPU
- Continuously running
- Scaling means intervention to add / remove servers



Amazon Lambda

- Virtual functions - no servers to manage!
- Limited by time - short executions
- Run on-demand
- Scaling is automated!

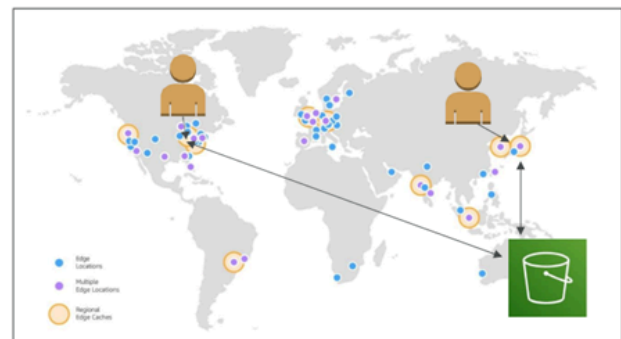
Benefits of AWS Lambda

- Easy Pricing:
 - Pay per request and compute time
 - Free tier of 1,000,000 AWS Lambda requests and 400,000 GBs of compute time
- Integrated with the whole AWS suite of services
- Event-Driven: functions get invoked by AWS when needed
- Integrated with many programming languages
- Easy monitoring through AWS CloudWatch
- Easy to get more resources per functions (up to 10GB of RAM!)
- Increasing RAM will also improve CPU and network!

Amazon CloudFront



- Content Delivery Network (CDN)
- Improves read performance, content is cached at the edge
- Improves users experience
- 216 Point of Presence globally (edge locations)
- DDoS protection (because worldwide), integration with Shield, AWS Web Application Firewall



Source: <https://aws.amazon.com/cloudfront/features/?nc=s&loc=2>

CloudFront - Origins

- S3 bucket
 - For distributing files and caching them at the edge
 - Enhanced security with CloudFront Origin Access Control (OAC)
 - OAC is replacing Origin Access Identity (OAI)
 - CloudFront can be used as an ingress (to upload files to S3)
- Custom Origin (HTTP)
 - Application Load Balancer
 - EC2 instance
 - S3 website (must first enable the bucket as a static S3 website)
 - Any HTTP backend you want

Redshift Overview



- Redshift is based on PostgreSQL, but it's not used for OLTP
- It's OLAP - online analytical processing (analytics and data warehousing)
- Load data once every hour, not every second
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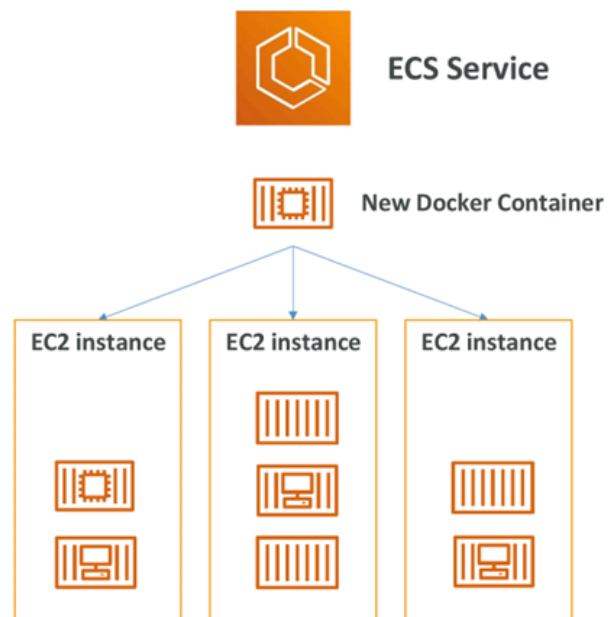
Amazon Kinesis



- For the exam: Kinesis = real-time big data streaming
- Managed service to collect, process, and analyze real-time streaming data at any scale
- Too detailed for the Cloud Practitioner exam but good to know:
 - Kinesis Data Streams: low latency streaming to ingest data at scale from hundreds of thousands of sources
 - Kinesis Data Firehose: load streams into S3, Redshift, ElasticSearch, etc...
 - Kinesis Data Analytics: perform real-time analytics on streams using SQL
 - Kinesis Video Streams: monitor real-time video streams for analytics or ML

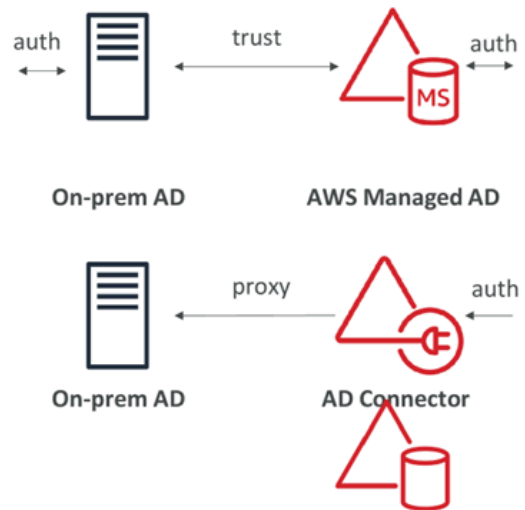
ECS

- ECS = Elastic Container Service
- Launch Docker containers on AWS
- You must provision & maintain the infrastructure (the EC2 instances)
- AWS takes care of starting / stopping containers
- Has integrations with the Application Load Balancer



AWS Directory Services

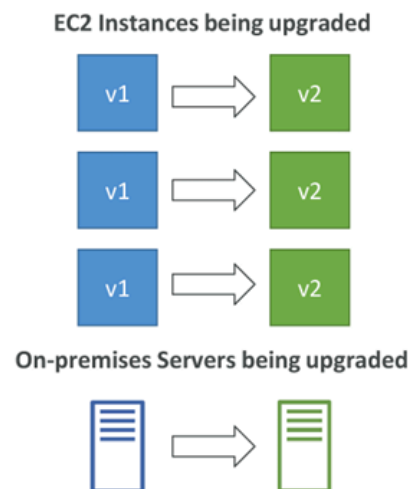
- AWS Managed Microsoft AD
 - Create your own AD in AWS, manage users locally, supports MFA
 - Establish "trust" connections with your on-premise AD
- AD Connector
 - Directory Gateway (proxy) to redirect to on-premise AD, supports MFA
 - Users are managed on the on-premise AD
- Simple AD
 - AD-compatible managed directory on AWS
 - Cannot be joined with on-premise AD



Devops(CI/CD)

AWS CodeDeploy

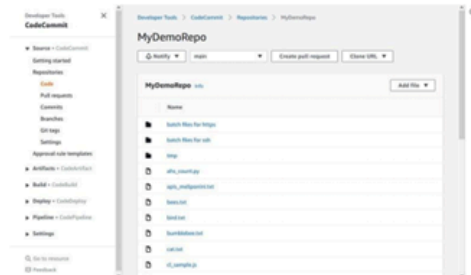
- We want to deploy our application automatically
- Works with EC2 Instances
- Works with On-Premises Servers
- Hybrid service
- Servers / Instances must be provisioned and configured ahead of time with the CodeDeploy Agent



AWS CodeCommit



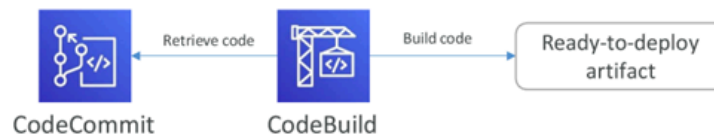
- Before pushing the application code to servers, it needs to be stored somewhere
- Developers usually store code in a repository, using the Git technology
- A famous public offering is GitHub, AWS' competing product is CodeCommit
- CodeCommit:
 - Source-control service that hosts Git-based repositories
 - Makes it easy to collaborate with others on code
 - The code changes are automatically versioned
- Benefits:
 - Fully managed
 - Scalable & highly available
 - Private, Secured, Integrated with AWS



AWS CodeBuild



- Code building service in the cloud (name is obvious)
- Compiles source code, run tests, and produces packages that are ready to be deployed (by CodeDeploy for example)

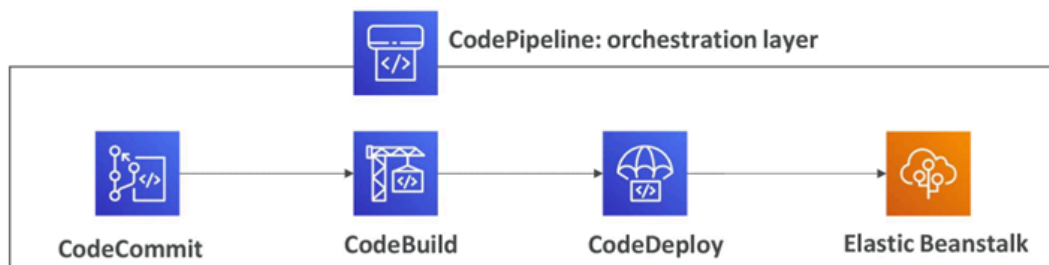


- Benefits:
 - Fully managed, serverless
 - Continuously scalable & highly available
 - Secure
 - Pay-as-you-go pricing - only pay for the build time

AWS CodePipeline



- Orchestrate the different steps to have the code automatically pushed to production
 - Code => Build => Test => Provision => Deploy
 - Basis for CI/CD (Continuous Integration & Continuous Delivery)
- Benefits:
 - Fully managed, compatible with CodeCommit, CodeBuild, CodeDeploy, Elastic Beanstalk, CloudFormation, GitHub, 3rd-party services (GitHub...) & custom plugins...
 - Fast delivery & rapid updates



12. Security on aws

AWS Shared Responsibility Model

- AWS responsibility - Security of the Cloud
 - Protecting infrastructure (hardware, software, facilities, and networking) that runs all the AWS services
 - Managed services like S3, DynamoDB, RDS, etc.
- Customer responsibility - Security in the Cloud
 - For EC2 instance, customer is responsible for management of the guest OS (including security patches and updates), firewall & network configuration, IAM
 - Encrypting application data
- Shared controls:
 - Patch Management, Configuration Management, Awareness & Training

Example, for S3



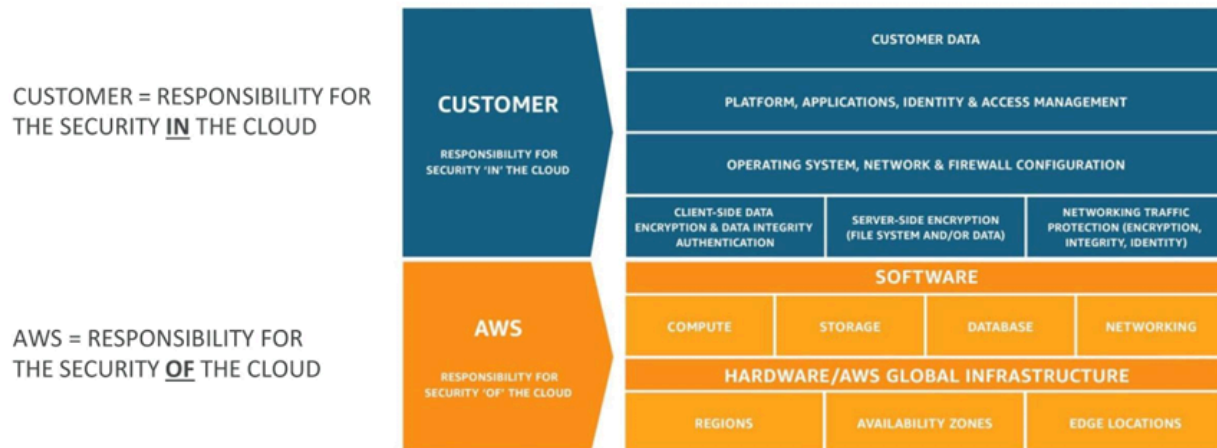
- AWS responsibility:
 - Guarantee you get unlimited storage
 - Guarantee you get encryption
 - Ensure separation of the data between different customers
 - Ensure AWS employees can't access your data
- Your responsibility:
 - Bucket configuration
 - Bucket policy / public setting
 - IAM user and roles
 - Enabling encryption

Example, for RDS

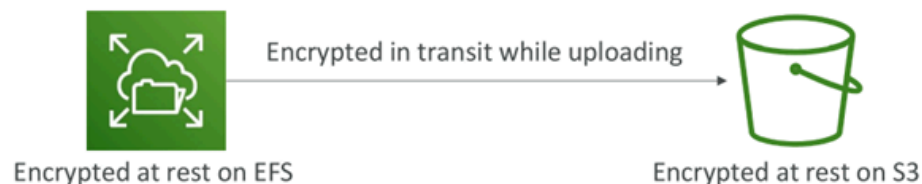


- AWS responsibility:
 - Manage the underlying EC2 instance, disable SSH access
 - Automated DB patching
 - Automated OS patching
 - Audit the underlying instance and disks & guarantee its functions
- Your responsibility:
 - Check the ports / IP / security group inbound rules in DB's SG
 - In-database user creation and permissions
 - Creating a database with or without public access
 - Ensure parameter groups or DB is configured to only allow SSL connections
 - Database encryption setting

Shared Responsibility Model diagram



Data at rest vs. Data in transit



- At rest: data stored or archived on a device
 - On a hard disk, on a RDS instance, in S3 Glacier Deep Archive, etc.
- In transit (in motion): data being moved from one location to another
 - Transfer from on-premises to AWS, EC2 to DynamoDB, etc.
 - Means data transferred on the network
- We want to encrypt data in both states to protect it!
- For this we leverage encryption keys

Root user privileges



- Root user = Account Owner (created when the account is created)
- Has complete access to all AWS services and resources
- Lock away your AWS account root user access keys!
- Do not use the root account for everyday tasks, even administrative tasks
- Actions that can be performed only by the root user:
 - Change account settings (account name, email address, root user password, root user access keys)
 - View certain tax invoices
 - Close your AWS account
 - Restore IAM user permissions
 - Change or cancel your AWS Support plan
 - Register as a seller in the Reserved Instance Marketplace
 - Configure an Amazon S3 bucket to enable MFA
 - Edit or delete an Amazon S3 bucket policy that includes an invalid VPC ID or VPC endpoint ID
 - Sign up for GovCloud

Additional services

Amazon Rekognition



- Find objects, people, text, scenes in images and videos using ML
- Facial analysis and facial search to do user verification, people counting
- Create a database of “familiar faces” or compare against celebrities
- Use cases:
 - Labeling
 - Content Moderation
 - Text Detection
 - Face Detection and Analysis (gender, age range, emotions...)
 - Face Search and Verification
 - Celebrity Recognition
 - Pathing (ex: for sports game analysis)

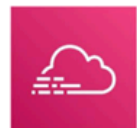
Amazon QuickSight



- Serverless machine learning-powered business intelligence service to create interactive dashboards
- Fast, automatically scalable, embeddable, with per-session pricing
- Use cases:
 - Business analytics
 - Building visualizations
 - Perform ad-hoc analysis
 - Get business insights using data
- Integrated with RDS, Aurora, Athena, Redshift, S3...

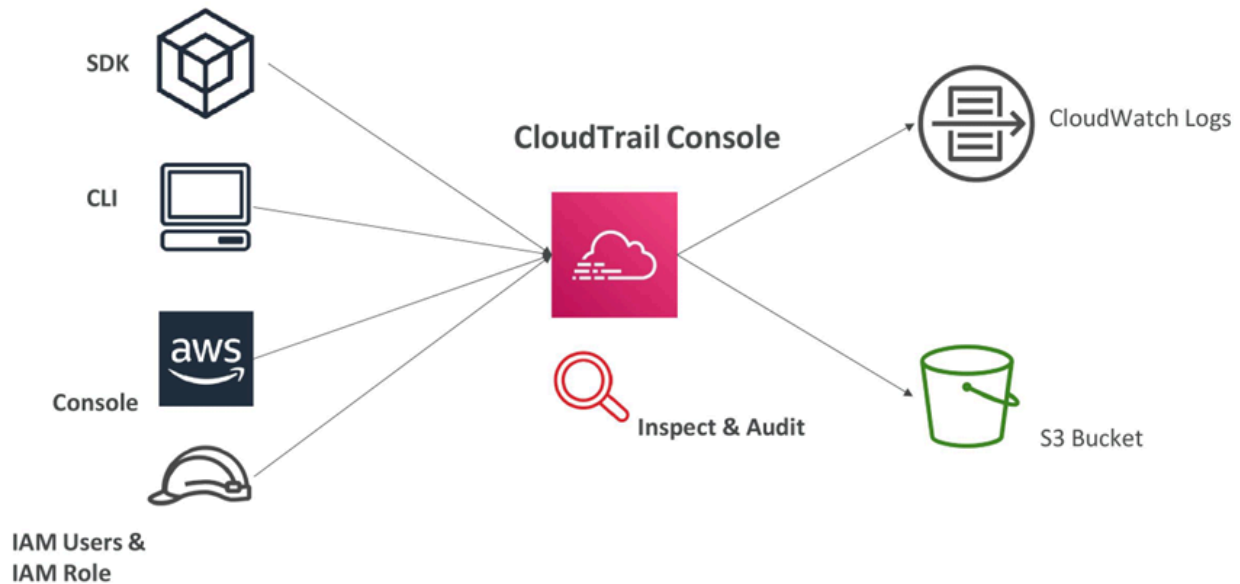
<https://aws.amazon.com/quicksight/>

AWS CloudTrail



- Provides governance, compliance and audit for your AWS Account
- CloudTrail is enabled by default!
- Get an history of events / API calls made within your AWS Account by:
 - Console
 - SDK
 - CLI
 - AWS Services
- Can put logs from CloudTrail into CloudWatch Logs or S3
- A trail can be applied to All Regions (default) or a single Region.
- If a resource is deleted in AWS, investigate CloudTrail first!

CloudTrail Diagram



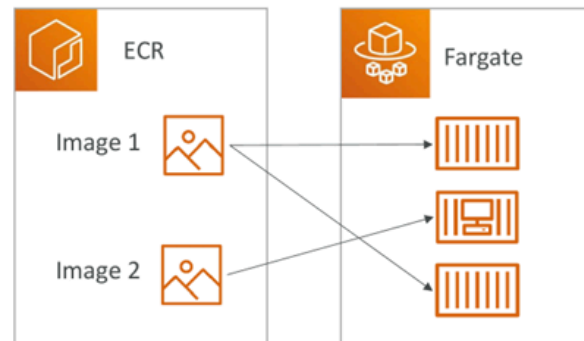
Amazon Lightsail



- Virtual servers, storage, databases, and networking
- Low & predictable pricing
- Simpler alternative to using EC2, RDS, ELB, EBS, Route 53...
- Great for people with little cloud experience!
- Can setup notifications and monitoring of your Lightsail resources
- Use cases:
 - Simple web applications (has templates for LAMP, Nginx, MEAN, Node.js...)
 - Websites (templates for WordPress, Magento, Plesk, Joomla)
 - Dev / Test environment
- Has high availability but no auto-scaling, limited AWS integrations

ECR

- Elastic Container Registry
- Private Docker Registry on AWS
- This is where you store your Docker images so they can be run by ECS or Fargate



AWS Marketplace



- Digital catalog with thousands of software listings from independent software vendors (3rd party)
- Example:
 - Custom AMI (custom OS, firewalls, technical solutions...)
 - CloudFormation templates
 - Software as a Service
 - Containers
- If you buy through the AWS Marketplace, it goes into your AWS bill
- You can sell your own solutions on the AWS Marketplace

AWS Support Plans Pricing



- Basic Support: free

Developer	Business	Enterprise On-Ramp	Enterprise
Greater of \$29.00	Greater of \$100.00	Greater of \$5,500.00	Greater of \$15,000.00
- or -	- or -	- or -	- or -
3% of monthly AWS charges	10% of monthly AWS charges for the first \$0-\$10K	10% of monthly AWS charges	10% of monthly AWS charges for the first \$0-\$150K
	7% of monthly AWS charges from \$10K-\$80K		7% of monthly AWS charges from \$150K-\$500K
	5% of monthly AWS charges from \$80K-\$250K		5% of monthly AWS charges from \$500K-\$1M
	3% of monthly AWS charges over \$250K		3% of monthly AWS charges over \$1M

AWS Ecosystem - AWS Support

DEVELOPER	• Business hours email access to Cloud Support Associates • General guidance: < 24 business hours • System impaired: < 12 business hours
BUSINESS	• 24x7 phone, email, and chat access to Cloud Support Engineers • Production system impaired: < 4 hours • Production system down: < 1 hour
ENTERPRISE	• Access to a Technical Account Manager (TAM) • Concierge Support Team (for billing and account best practices) • Business-critical system down: < 15 minutes

Your AWS Certification journey

FOUNDATIONAL

Six months of fundamental AWS Cloud and industry knowledge



PROFESSIONAL

Two years of experience designing, operating, and troubleshooting solutions using the AWS Cloud



ASSOCIATE

One year of experience solving problems and implementing solutions using the AWS Cloud



SPECIALTY

Technical AWS Cloud experience in the Specialty domain as specified in the exam guide

